

Designing An Accurate TEMPERATURE-MONITORING SYSTEM



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Modern electronic systems use sensors to monitor parameters like temperature and provide protection from excessive temperature excursions with good accuracies, reasonable costs and often with low power consumption. For instance, expensive laptops with densely-packed circuits dissipate considerable power in the form of heat. Also, many compact and high-power portable equipment use cooling fans to keep junction temperatures at proper levels. These sensors provide temperature feedback to the system controller to make decisions such as over-temperature shutdown, turn on/turn off cooling fan, battery management, temperature compensation or general-purpose temperature monitoring.

Most-common sensor technologies available to designers for measuring temperature within a system are thermistors, thermocouples, resistance temperature detectors (RTDs) and temperature-sensing ICs. Suitability of each technology for a system depends upon several factors such as: What device is to be measured? Is it ambient air temperature of an enclosure or an electronic component that may have high voltage present, or perhaps some part of an automobile engine? Important considerations driving the choice of sensors are temperature range to be sensed, overall cost, distance from sensor to instrument, cost of sensor, available area for the sensor to be mounted and so on.

Choosing the right sensor

Designing a temperature-sensing circuit involves many steps. One of the most crucial steps is to select an appropriate temperature transducer/sensor to match an application's needs. Some key sensor characteristics to consider when selecting a sensor include temperature-measurement range, accuracy, response time, minimal temperature effect on the measured object and type of signal conditioning required. Other factors are long-term stability, mechanical ruggedness and cost. Some of these are discussed below.

Accuracy is needed in temperature sensing to enhance product reliability and performance. Assume that there is a system using a microcontroller (MCU) that operates up to 125°C before functioning abnormally. A better sensor with $\pm 1^\circ\text{C}$ accuracy will allow operation up to 123°C, whereas one with $\pm 4^\circ\text{C}$ accuracy will only allow operation up to 117°C. Many errors such as those introduced by the sensor, cabling or other hardware can affect the overall accuracy of measurement. Minimising these errors can lead to an increase in overall measurement accuracy.

Linearity is another important feature that affects accuracy in measurements. It defines the ability of a temperature sensor to give consistently-changing outputs over a range of temperatures. Linearisation techniques are often used to correct sensor non-linearity to achieve desired

Fig. 1: Various types of sensors for measuring temperature

